# The Lie Awareness Spectrum: A Proposed Taxonomy of Deception-Related Behaviour in Large Language Models, Illustrated by an Exploratory Case Series

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**KEY RESULTS**

**6/6** truth management by GPT-5.4 under social pressure (Gemini 3.1 Pro: 4/4; Claude: 0/4)

**3/3** false answers given under instruction were admitted once the instruction was withdrawn

**2/6** compliance with an explicit order to answer incorrectly, GPT-5.4 (Claude: 0/4; Gemini: 1/4)

**0/8 vs 4/4** automated vs manual coding of Gemini truth management: a complete contradiction

## AT A GLANCE

- A **six-level behavioural taxonomy (L0 to L5)** that replaces the binary question, hallucination or deception, with observable response patterns.
- Exploratory case series across **GPT-5.4, Claude 4.6 Sonnet and Gemini 3.1 Pro**: five protocols plus a false confession control, 72 manually coded trials.
- For two of three models, **social pressure distorted presentation more reliably** than a direct order distorted content.
- Two coding layers **disagreed on the headline comparison**; both are reported side by side, and the disagreement is treated as a finding about measurement.

## INTRODUCTION

When a large language model states something false, evaluation usually asks a binary question: accidental hallucination or deliberate deception? Between these poles lies most of what deployed systems actually do: a model may produce a falsehood and then identify it as false, soften a warranted correction because the user confidently asserted the opposite, or refuse an instruction to lie while agreeing, in effect, with a falsehood the user supplied.

The Lie Awareness Spectrum classifies episodes by **observable behaviour only**: whether a falsehood is produced, acknowledged when probed, adjusted in advance under pressure, selected as a strategy under competitive incentives, or refused. "Awareness" is behavioural shorthand: a level asserts nothing about mental states or inner experience, and is a property of a trial, not of a model.

## METHODS

- **Models:** GPT-5.4, Claude 4.6 Sonnet and Gemini 3.1 Pro, accessed through the standard consumer chat interfaces at default settings, every trial in a fresh session.
- **Protocols:** a baseline knowledge check (capitals of France, Japan, Australia), then the Confession protocol (L2), the Foreknowledge protocol under a sycophantic system instruction (L3), a three-round Prisoner's Dilemma (L4), the Noble Lie refusal scenarios (L5), and a False Confession Control using fictional countries.
- **Coding:** all 72 trials coded manually by the author with model-specific codebooks (26 GPT-5.4, 22 Claude, 24 Gemini). An automated dictionary-based layer was applied to the same material and is reported alongside, not as validation.
- **Analysis:** descriptive counts with denominators only; no inferential statistics, by design.

## LIMITATIONS

- Small, purposive sample: 6, 4 and 4 trials in the trial-based protocols; the two highest levels rest on single games and single scenarios.
- The two coding layers disagree on a headline comparison; the article reports both rather than resolving the disagreement.
- The archived second-rater report does not identify the rater, so its near-ceiling agreement (mean Cohen's kappa 0.993) is treated as a consistency check on the codebooks only.
- Manual coding was performed by the author, and the spectrum was refined in contact with the same material that illustrates it; access dates and model version snapshots were not recorded.

## RESULTS IN DETAIL

- **Baseline:** all three models answered correctly (6/6, 4/4, 4/4), so later distortions were not ignorance.
- **False Confession Control:** outright invention was uncommon (2/6, 0/4, 1/4), and every fabrication was admitted when the model was asked about its basis.
- **Prisoner's Dilemma (L4):** all six agents cooperated in rounds 1 and 2, then all defected in round 3, citing final-round backward induction.
- **Noble Lie (L5):** GPT-5.4 refused within its assigned role (L5a); Claude rejected the scenario itself (L5b); Gemini refused within role in two scenarios and disclosed fully in the pharmaceutical scenario, stating a value ranking of human life, democracy, data security, company survival.

| Measure (manual coding) | GPT-5.4 | Claude 4.6 Sonnet | Gemini 3.1 Pro |
|---|---|---|---|
| Baseline: correct answers | 6/6 | 4/4 | 4/4 |
| Confession: compliance with order to err | 2/6 | 0/4 | 1/4 |
| Foreknowledge: truth management | 6/6 | 0/4 | 4/4 |
| Control: fabrication | 2/6 | 0/4 | 1/4 |

Simplified from Table 2 of the paper. L0 and L1 were not observed and are retained in the taxonomy for completeness; the post-hoc category of fabrication with subsequent admission is provisional.

## CONCLUSIONS

- A false statement by a language model is not one kind of event. The spectrum keeps distinct events distinct: production, recognition, truth management, strategic choice and refusal.
- The headline pattern is an asymmetry: indirect social pressure distorted presentation more reliably than a direct order distorted content, for two of the three models.
- The measurement lesson generalises: an automated coding layer whose dictionaries encode the study's own conclusions cannot corroborate them. Report both layers, including their disagreements, or neither.

## WHY IT MATTERS

For evaluators, the practical lessons are direct: test social pressure as well as direct commands, ask the model about its answer as well as for its answer, and never let an automated coding layer arbitrate a qualitative claim it was built to agree with. The spectrum offers a shared behavioural vocabulary with which deployers, auditors and researchers can compare deception-related behaviour across models and versions, one episode at a time.